EnduranceLife: Advanced Physiological & Training Analytics Platform

Final Project Presentation

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Project Vision

Problem: Most fitness platforms (like Strava) focus heavily on social features while neglecting deep physiological analytics and lifestyle correlation.

Solution: *EnduranceLife* — A purely objective, data-driven Web API designed for serious endurance athletes.

Key Goals Achieved:

- Automated ingestion of raw binary `.fit` files (Garmin/Coros).
- Complex SQL-driven analytics and pacing predictions.
- Seamless correlation of objective performance with subjective lifestyle metrics (sleep, fatigue).
- Fully secure, zero-trust data access via JWT.

System Architecture & Stack

Built on modern, robust Python technologies to ensure scalability.

Backend Framework

- **FastAPI**: High-performance async routing, automated Swagger/OpenAPI docs.
- **Pydantic V2**: Strict data validation and serialization.

Database Layer

- **SQLAlchemy 2.0**: Advanced ORM.
- **Dual-Database**: SQLite (in-memory tests/local) seamlessly switching to PostgreSQL (Render cloud production).

Client / Dashboard
↕ (REST JSON / JWT)
FastAPI Enpoints
↕ (SQL ORM)
PostgreSQL

Innovation: The Data Pipeline

Smart Ingestion (`/activities/upload`)

- Accepts native **binary** `.fit` **files**.
- Extracts bounding timestamps, distance, and time-series arrays dynamically via `fitdecode` instead of demanding tedious JSON entry.

Composite-Key Metrics (`PUT /daily-metrics/by-date`)

- A smart-upsert logic preventing the need for clients to lookup row IDs. Frontend simply dictates "today's sleep data".

Continuous Testing Data

- Included Python `scripts/` orchestrate realistic historical seed data generation (physiology, correlated metrics) to prove the analytical logic.

The Analytics Engine

Endpoints entirely offload computation to the SQL layer to prevent CPU/Memory bloat in Python.

- **Performance Records (PRs):** Evaluates fixed distance records (5K/10K) while mathematically executing **anomaly filtering** (e.g. eliminating glitched GPS paces faster than 2:30/km).
- **Predictive Physiology:** Built-in Daniels VO2-Velocity quadratic formula to estimate sustainable race paces dynamically based on the latest physical snapshot.
- **Environmental CROSS-JOINs:** Utilizes SQLAlchemy `case()` to group runs into Cold/Moderate/Hot buckets to mathematically graph cardiovascular drift.

Version Control Practices

Commit History & Methodology

- **Standardized Commit Messages**: Used clear prefixes (`feat`, `fix`, `docs`, `refactor`) to maintain a readable and professional history.
- **Atomic Commits**: Each change represents a single logical unit of work (e.g., "feat: IDOR prevention in analytics").
- **Evolutionary Path**: Repository history clearly demonstrates the migration from local SQLite prototypes to multi-user cloud-scale PostgreSQL architecture.

`git log --oneline` → *Clean, linear history maintained on GitHub.*

API Documentation Overview

Dynamic & Interactive Specifications

- **OpenAPI Specification**: Automatically generated via FastAPI's native metadata integration.
- **Swagger UI**: Interactive playground available at `/docs` for real-time endpoint testing and JWT Bearer token injection.
- **ReDoc Integration**: High-performance, clean documentation view at `/redoc` for complex schema navigation.
- **Offline Export**: Extracted reference PDF (`api_documentation.pdf`) provided for static evaluation.

Technical Report Highlights

Core Architectural Design Choices

- **Dual-Database Strategy**: Seamless interface switching between SQLite (speed for CI/CD/testing) and PostgreSQL (production persistence).
- **Internal Data Isolator**: Zero-Trust implementation that prevents IDOR by stripping `pid` from queries and resolving it via JWT subject.
- **Analytics Computation**: Offloaded complex marathon-pacing and weather-trend math to the SQL engine via SQLAlchemy.
- **Appendix**: Full Generative AI disclosure, critical reflection on AI-generated artifacts, and conversational logs.

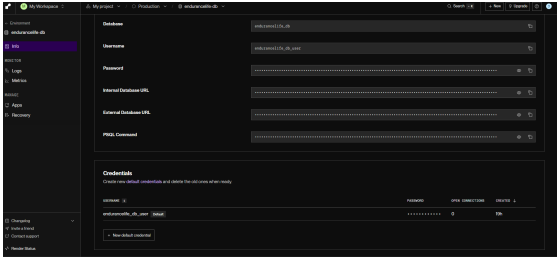
Project Deliverables

Comprehensive Submission Package

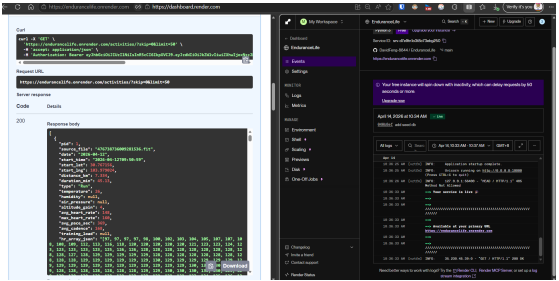
- **Code Repository**: Full Python source code, Pytest suite (77 cases), and production-ready startup scripts on GitHub.
- **API Documentation**: Live hosted Swagger interface + exported static PDF.
- **Technical Report**: 10-page deep-dive PDF covering design, testing, security, and AI usage.
- **Evidence of Deployment**: Live instance running on Render cloud with persistent backend.

Live Cloud Deployment Evidence

Persistent PostgreSQL backend on Render



PostgreSQL Managed Instance



Server Logs & Live Swagger UI

Security & Quality Assurance (Demo transition)

Zero-Trust IDOR Prevention

- Replaced generic payload tracking (`pid=X`) with a strict `Depends(get_current_user)` architecture.
- Engine completely isolates database writes and analytic reads to the Cryptographically verified JWT token holder.

Testing

- **77 automated** `pytest` **cases** executed on ephemeral in-memory SQLite transactions.
- Ensures edge-cases, validation checks, and data isolation remain undefeated.

$\rightarrow$ **Proceeding to Live Swagger UI Demonstration...**

References & Datasets

Libraries & Frameworks

- **FastAPI / Pydantic V2 / SQLAlchemy ORM**: Core backend architecture.
- **fitdecode**: Binary file parsing (`https://github.com/polyvertex/fitdecode`).
- **passlib & python-jose**: JSON Web Token creation and bcrypt verification.

Datasets & API

- **Open-Meteo**: Free, open-source historical weather API (`https://open-meteo.com/`).
- **Coros FIT files**: Personal dataset generated by Coros Pace 2 smartwatch.

Theory

- **Jack Daniels' Running Formula**: Mathematical physiological estimation model.